

# MAYA BOSE

## EDUCATION

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| <b>University of Michigan, Ann Arbor</b><br><i>PhD, Computational Medicine and Bioinformatics</i> | Aug. 2021– May 2026 |
| <b>University of Arizona</b><br><i>BS with Honors, summa cum laude, Bioinformatics</i>            | June 2021           |

## RESEARCH EXPERIENCE

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| <b>Graduate Student Research Assistant</b>   <i>PI: Dr. Stephen Parker</i><br>University of Michigan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | March. 2022 – Present |
| <ul style="list-style-type: none"><li>Performed randomized batch sorting, QC, clustering, differential cell type abundance analysis, differential gene expression analysis, and differential chromatin accessibility of snRNAseq and snATACseq multiome libraries for HFpEF study in collaboration with Pfizer and Wake Forest University</li><li>Benchmarked cutting-edge differential gene expression tools MAST and miloDE against DESeq2 to identify optimal data analysis strategies for my dataset</li><li>Automated reproducible workflows using Snakemake, Nextflow, and Singularity</li><li>Predicted the expression of CAGE-derived TSS in a massively parallel reporter assay from tissue-specific regulatory annotations using generalized linear models and elasticnet regression</li></ul> |                       |
| <b>Graduate Rotation Student</b>   <i>PI: Ryan Mills</i><br>University of Michigan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jan. 2022 – Mar. 2022 |
| <b>Graduate Rotation Student</b>   <i>PI: Lydia Freddolino</i><br>University of Michigan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Oct. 2021 – Dec. 2021 |
| <b>Undergraduate Researcher</b>   <i>PI: Rebecca Mosher</i><br>University of Arizona                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | May 2019 – May 2021   |
| <b>Undergraduate Researcher</b>   <i>PI: Aurelio José Figueredo</i><br>University of Arizona                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Jan. 2018 – May 2021  |

## PUBLICATIONS

- [1] **M. Bose**, A. Varshney, M. Piper, E. Pashos, V. Dolgachev, D. Kitzman, E. B. Fauman, F. Damilano, R. J. Roth Flach, B. Nicklas, and S. C. Parker. "Single-Nucleus Multiome Profiling in Human Skeletal Muscle Reveals Differential Gene Expression, Cell Type Abundance, and Chromatin Accessibility in Heart Failure with Preserved Ejection Fraction". In: (In preparation).
- [2] **M. Bose**, A. Tovar, Y. Kyono, A. Varshney, J. O. Kitzman, and S. C. J. Parker. "Massively Parallel Reporter Assay Reveals Promoter-, Position-, and Strand-Specific Effects in Transcription Start Sites". In: *bioRxiv : the preprint server for biology* (Oct. 14, 2025). doi: 10.1101/2025.10.13.659964.
- [3] A. Tovar, Y. Kyono, K. Nishino, **M. Bose**, A. Varshney, S. C. J. Parker, and J. O. Kitzman. "Using a Modular Massively Parallel Reporter Assay to Discover Context-Specific Regulatory Grammars in Type 2 Diabetes". In: *bioRxiv : the preprint server for biology* (2023). doi: 10.1101/2023.10.08.561391.

## ORAL PRESENTATIONS

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- [1] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". NHGRI Research Training & Career Development Meeting (Seattle, WA). Apr. 2024.
- [2] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". American Society of Human Genetics Annual Meeting (Los Angeles, CA). Oct. 2022.
- [3] "RNA-Directed DNA Methylation (RdDM)". Undergraduate Biology Research Program Summer Presentation (Tucson, AZ). Aug. 2020. URL: <https://youtu.be/MVFeofaltjs>.

## POSTER PRESENTATIONS

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- [1] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". NHGRI Research Training & Career Development Meeting (Philadelphia, PA). Apr. 2025.
- [2] "Single-Nucleus Multiome Profiling in Human Skeletal Muscle Reveals Differential Expression and Cell Type Abundance in Heart Failure with Preserved Ejection Fraction". Accelerating Medicines Partnership® Program – Common Metabolic Diseases Parliament Meeting (Philadelphia, PA). Apr. 2025.
- [3] "Single-Nucleus Multiome Profiling in Human Skeletal Muscle Reveals Differential Gene Expression, Cell Type Abundance, and Chromatin Accessibility in Heart Failure with Preserved Ejection Fraction". American Society of Human Genetics Annual Meeting (Boston, MA). Oct. 2025.
- [4] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". American Diabetes Association Annual Meeting (Orlando, FL). June 2024.
- [5] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". NHGRI Research Training & Career Development Meeting (Salt Lake City, UT). Apr. 2023.
- [6] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". Department of Computational Medicine and Bioinformatics Annual Retreat Poster Session (Ann Arbor, MI). Sept. 2023.
- [7] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". Department of Human Genetics and Department of Ecology Joint Retreat Graduate Student Poster Session (Ann Arbor, MI). Oct. 2023.
- [8] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". American Society of Human Genetics Annual Meeting (Washington, D.C.). Nov. 2023.
- [9] "Massively Parallel Reporter Assay Reveals Promoter-, Position-Dependent and Tissue-Specific Effects in Islet TSSs". DCMB 10th Anniversary Symposium Graduate Student Poster Session (Ann Arbor, MI). Oct. 2022.
- [10] "23-Nt sRNAs with Terminal 3' Untemplated Nucleotides Form a Small Subset of RDR2 Transcription Products". Undergraduate Biology Research Program Virtual Poster Conference (Tucson, AZ). Jan. 2020.
- [11] "Abundances of RdDM-Produced 23nt and 24nt sRNAs Vary Across Tissue Types in Brassica Rapa and Arabidopsis Thaliana". Undergraduate Biology Research Program Virtual Poster Conference (Tucson, AZ). Jan. 2019.

## TEACHING EXPERIENCE

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**Girls Who Code Guest Speaker**  
University of Michigan

Sept. 24, 2024

- Tutored undergraduate students in College Algebra, Pre-Calculus, and Calculus
- Obtained Level I International Tutor Training Program Certification

## SCHOLARSHIPS AND AWARDS

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| <b>Rackham Conference Travel Grant</b>   <i>University of Michigan</i>                                                         | Oct. 2025             |
| <b>Excellence in Leadership Award Department Nominee</b>   <i>University of Michigan Bioinformatics</i>                        | Mar. 2025             |
| <b>Rackham Conference Travel Grant</b>   <i>University of Michigan</i>                                                         | Oct. 2023             |
| <b>Bioinformatics Student Service Award</b>   <i>University of Michigan Bioinformatics</i>                                     | Oct. 2023             |
| <b>Rackham Conference Travel Grant</b>   <i>University of Michigan</i>                                                         | Oct. 2022             |
| <b>Genome Sciences Training Program T32</b>   <i>National Human Genome Research Institute</i>                                  | July 2022 – July 2025 |
| Normally a 2-year grant, was awarded a 3rd year of support in recognition of scientific achievement and service to the program |                       |
| <b>Benard Maas Fellowship</b>   <i>University of Michigan</i>                                                                  | May. 2021             |
| <b>Da Vinci Award</b>   <i>University of Arizona Honors College</i>                                                            | Mar. 2021             |
| <b>Excellence in Undergraduate Research Award</b>   <i>University of Arizona Ecology and Evolutionary Biology</i>              | Mar. 2021             |
| <b>Honors Legacy Award</b>   <i>University of Arizona</i>                                                                      | Feb. 2020             |
| <b>Galileo Circle Scholarship</b>   <i>University of Arizona</i>                                                               | Apr. 2020             |
| <b>Wildcat Excellence Scholarship</b>   <i>University of Arizona</i>                                                           | Aug. 2021 – May 2021  |

## SERVICE AND OUTREACH

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| <b>Bioinformatics Graduate Student Association Executive Member</b><br>University of Michigan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Jan. 2022 – Present |
| <ul style="list-style-type: none"><li>• Organize monthly community-building social events for graduate students. These events regularly draw 25-40 students in attendance</li><li>• Facilitate communication between the graduate students and faculty through town hall discussions and placement of students on department committees</li><li>• Identify and recruit students for placement in student organizations and department committees</li><li>• Promote the retention of a diverse cohort of graduate students by fostering an inclusive and supportive academic environment</li></ul> |                     |
| <b>UM Bioinformatics &amp; PIBS Recruitment Host</b><br>University of Michigan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Jan. 2022 – Present |
| <ul style="list-style-type: none"><li>• Host prospective graduate students across both virtual and in-person recruitment weekends</li><li>• Fulfilled duties as lead host for 2023, 2024. Served as point person during interview weekends in addition to regular responsibilities</li></ul>                                                                                                                                                                                                                                                                                                      |                     |
| <b>DCMB Pi Day Potluck Committee Member</b><br>University of Michigan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2025                |
| <b>DCMB Retreat Committee Member</b><br>University of Michigan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2023; 2024          |

**DCMB Yearly Summer Picnic Committee Member**

University of Michigan

2023

**UA COVID-19 Vaccine POD Volunteer**

University of Arizona

Mar. 2021 – May 2021

**UA Ecology & Evolutionary Biology Department Ambassador**

University of Arizona

Jan. 2020 – May 2021